Journal Name

Crossmark

RECEIVED
dd Month yyyy

REVISED
dd Month yyyy

PAPER

Deployment-Realistic Performance Measurement of Low-Cost MEMS Sensing for IoT-Based Condition Monitoring of Induction Machines

Author list removed for anonymity

E-mail:

Keywords: condition monitoring, fault diagnosis, induction machines, MEMS accelerometer, sensor characterisation, measurement uncertainty, performance measurement, Internet of Things, edge computing, low-cost sensing

Abstract

Vibration-based condition monitoring on low-cost sensors hinges on two measurement questions that are often conflated: how much fault-relevant information a consumer-grade micro-electro-mechanical-systems (MEMS) accelerometer can actually measure, and how the diagnostic performance of such a sensor should itself be measured. Both are addressed for a smartphone-class accelerometer applied to induction-machine condition monitoring, using a publicly available 100 Hz vibration dataset of six health states, three supply frequencies, and two load conditions. The sensor is first characterised as a measurement instrument: its effective resolution (≈ 0.015 mg), a broadband signal floor (-70.8 dB re 1 g^2/Hz that upper-bounds the sensor noise), and usable band—bounded by the 50 Hz Nyquist limit—are established from the data, and an information-theoretic channel analysis shows that the on-device gravity compensation preserves the vibration power but reshapes its low-frequency structure, leaving the raw acceleration channels measurably more informative. A leakage-free, recording-wise protocol is then introduced to obtain unbiased performance estimates; it reveals that the near-perfect accuracies reported under conventional random-split evaluation are a measurement artefact of window overlap and recording identity, overstating accuracy by 41 to 47 percentage points. Reported with expanded uncertainty, an accuracy–resource characterisation shows that a resource-frugal configuration—25 Hz sampling, three raw channels, and a 4 s window—classifies the health state more accurately than the full-rate configuration (a paired improvement of 23 percentage points; paired t-test $p = 0.0001$) while reducing the data volume roughly eight-fold, which in turn defines low-bandwidth operating points for IoT nodes over Bluetooth Low Energy, Zigbee, NB-IoT, and LoRaWAN. The measurement protocol set out here, reported in full, provides a reference for the deployment-honest evaluation of low-cost vibration-sensing systems.

1 Introduction

Induction machines account for a dominant share of the electrical energy consumed by industry and form the mechanical backbone of countless processes; their unexpected failure causes costly downtime and, in critical applications, safety hazards. Continuous condition monitoring (CM) and early fault diagnosis are therefore central to modern maintenance practice [1, 2]. Among the available modalities, vibration analysis is the most established, because the mechanical and electrical faults of interest—bearing defects, rotor asymmetries, and supply imbalances—imprint characteristic signatures on the machine's vibration response [3, 4].

The prevailing paradigm relies on industrial-grade piezoelectric accelerometers sampled at tens to hundreds of kilohertz, which resolve the high-frequency bearing defect frequencies exploited by spectral and envelope analysis. This is reflected in the public benchmarks that anchor the field, acquired at 8–200 kHz [5, 6, 7, 8]. Although accurate, such instrumentation is costly, requires intrusive installation, and scales poorly to the vast population of small, low-value machines that individually do not justify a dedicated monitoring channel yet collectively dominate a plant's machine inventory. Closing this coverage gap motivates a shift toward inexpensive, wireless, and readily deployable sensing.

Consumer micro-electro-mechanical-systems (MEMS) inertial sensors, epitomised by those embedded in smartphones, offer such a platform at negligible marginal cost [9]. Machine fault detection has been demonstrated from smartphone- and MEMS-acquired vibration [10, 11, 12], and dedicated low-cost MEMS nodes now integrate on-board processing with wireless telemetry for machine diagnostics [13, 14]. These sensors, however, operate at sampling rates one to three orders of magnitude below industrial practice—100 Hz in the present case—so that, by the Nyquist criterion [15, 16], the high-frequency signatures underpinning classical bearing diagnosis are not observable. A fundamental and largely unresolved question follows: what fault information is genuinely recoverable from such a constrained sensor, and does the apparent performance transfer to a real deployment?

The second half of that question is easily overlooked. Learning-based fault diagnosis routinely reports near-perfect accuracy, yet a growing body of evidence attributes much of this success to *data leakage*—information shared between training and test partitions that inflates measured accuracy without improving generalisation [17]. In vibration diagnosis, leakage arises when overlapping analysis windows from a single continuous recording are divided across partitions, or when recording-specific characteristics are learned in place of fault features; enforcing component- or recording-wise separation has been shown to reduce reported accuracy substantially [18, 19]. This scrutiny has so far been directed at industrial-grade, high-rate data. For low-rate smartphone datasets—in which each operating condition is captured in one continuous acquisition and heavy window overlap is common—the risk is arguably greater, yet random-split evaluation remains the norm, including in the baseline accompanying the dataset examined here [20].

A third consideration is deployment. The Industrial Internet of Things envisions dense networks of inexpensive nodes that stream, or locally distil, diagnostic information over constrained wireless links [21], and recent TinyML implementations have shown that lightweight classifiers can run on microcontrollers at millisecond latency and sub-millijoule energy [22, 23]. Such work establishes that edge diagnosis is feasible, but it generally presupposes industrial-bandwidth sensing and does not characterise the coupled trade-off between diagnostic accuracy and the sampling, communication, and computational cost that governs an ultra-low-rate node. Whether, for a smartphone-class sensor, the low sampling rate is a liability to be tolerated or an asset to be exploited has not been quantified.

This paper approaches these gaps as a measurement problem. Using a public smartphone-acquired vibration dataset of an induction machine recorded across six health states, three supply frequencies, and two load conditions at 100 Hz [20, 24], and rather than proposing a new classifier, the study foregrounds measurement rigour: the consumer sensor is characterised as a measurement instrument, a protocol is developed for obtaining unbiased and uncertainty-quantified estimates of its diagnostic performance, and the accuracy–resource trade-off that governs an IoT deployment is mapped. Two measurement questions organise the work—how much fault information the sensor can convey, and how that performance must be measured if it is to transfer to deployment. The principal contributions of this paper are fourfold. First, it introduces a *performance-measurement methodology* for windowed low-rate vibration condition monitoring—comprising leakage-free recording-wise and cross-condition (cross-speed and cross-load) protocols with paired significance testing and expanded-uncertainty reporting—and applies it to quantify the measurement bias introduced by the prevailing random-split practice, which is shown to overstate accuracy by approximately 41 to 47 percentage points. Second, it provides a *measurement characterisation* of the smartphone MEMS channel that establishes, with uncertainty, its effective resolution, signal floor, and usable band, and reveals that the on-device gravity compensation preserves the vibration power but reshapes its low-frequency structure, leaving the raw channels the more informative and hence preferable for condition monitoring. Third, it develops an *accuracy–resource characterisation* that identifies a resource-frugal operating point which, under leakage-free evaluation, classifies the machine's health state significantly more accurately than the full-rate configuration, and it translates this trade-off, through a communication link budget and an edge-model footprint, into deployable operating points for Bluetooth Low Energy, Zigbee, NB-IoT, and LoRaWAN nodes. Finally, it establishes an *uncertainty-aware reference baseline* for low-cost IoT-based induction-machine condition monitoring, with the feature set, evaluation protocols, estimator settings, and random seeds reported in full, so that the baseline can be reimplemented independently.

The remainder of this paper is organised as follows. Section 2 reviews related work on vibration-based condition monitoring, low-cost MEMS sensing, IoT and edge diagnosis, and evaluation methodology. Section 3 describes the dataset and measurement setup, and Section 4 characterises the sensor and its signals. Section 5 details the processing chain and the evaluation protocols. Section 6 reports the results, Section 7 discusses their implications and limitations, and Section 8 concludes.

2 Related Work

2.1 Vibration-Based Condition Monitoring of Induction Machines

Vibration analysis is a mature and widely adopted modality for detecting mechanical and electrical faults in induction machines, including bearing defects, rotor asymmetries, and supply imbalances [1, 2]. Classical pipelines rely on industrial-grade piezoelectric accelerometers sampled at tens to hundreds of kilohertz, which resolve characteristic bearing defect frequencies—ball-pass frequency outer race (BPFO), ball-pass frequency inner race (BPFI), and ball-spin frequency (BSF)—and their harmonics through spectral and envelope analysis. This paradigm underpins the public benchmarks that dominate the field: representative datasets are acquired at 8–16 kHz [5], 42 kHz [6], 51.2 kHz [7], and up to 200 kHz [8]. More recently, deep learning has become the default classifier family for such signals [3, 4]. High-frequency acquisition, however, entails costly instrumentation, intrusive cabling, and limited portability, which restrict deployment to the comparatively small population of critical machines and leave the many low-value machines in a plant unmonitored.

2.2 Low-Cost MEMS and Smartphone-Based Vibration Sensing

Micro-electro-mechanical-systems (MEMS) inertial sensors have emerged as an inexpensive, compact alternative for vibration acquisition. Their metrological limits relative to reference-grade accelerometers—narrower bandwidth and higher noise floor—are well documented [9], and condition-monitoring practitioners have generally regarded consumer-grade MEMS bandwidth as insufficient for bearing diagnosis [13]; the metrological variability of such low-cost sensors and its effect on diagnostic features has itself become a subject of measurement research [25]. Nonetheless, a growing body of work demonstrates fault detection from smartphone- or MEMS-acquired signals, using mobile applications and, more recently, deep networks [10, 11, 12]. In parallel, purpose-built low-cost MEMS sensor nodes combine on-node processing with short-range or long-range radios for machine diagnostics [13, 14]. The dataset employed in this study [20, 24] extends this line by providing long-duration, multi-condition smartphone-MEMS recordings at a software-capped 100 Hz. Crucially, the accompanying baseline reports near-perfect classification accuracy under a random split of overlapping windows, a protocol that is scrutinised in Section 2.4.

2.3 IoT, Edge, and TinyML for Condition Monitoring

The Industrial Internet of Things reframes condition monitoring as a distributed sensing problem in which many inexpensive nodes stream—or locally distil—diagnostic information over constrained wireless links [21]. Because raw high-rate vibration cannot be sustained over low-power wide-area radios, a large fraction of recent work pushes feature extraction or inference onto the node itself. TinyML implementations of lightweight classifiers on microcontrollers report bearing-fault accuracies with millisecond-scale latency and sub-millijoule energy per inference [22, 23], and low-cost MEMS nodes integrate on-board spectral analysis with LoRa or Bluetooth Low Energy telemetry [14], and smart-sensor systems combine vibration measurement with on-board bearing diagnosis [26]. These studies establish the feasibility of edge diagnosis but typically assume industrial-bandwidth sensing and rarely quantify the coupled trade-off between sampling/communication cost and diagnostic accuracy that governs an ultra-low-rate node.

2.4 Evaluation Methodology and Data Leakage

A recurring threat to the credibility of learning-based fault diagnosis is data leakage—the inadvertent sharing of information between training and test partitions that inflates reported accuracy and does not transfer to deployment [17]. In vibration diagnosis specifically, two mechanisms dominate: (i) splitting *overlapping* analysis windows drawn from the same continuous recording, so that near-duplicate segments populate both partitions; and (ii) condition- or recording-wise correlations that let a model recognise *which recording* a window originates from rather than the fault itself. Recent studies show that enforcing component-wise (e.g., bearing-wise) or recording-wise separation causes headline accuracies to fall substantially [18, 19]. Despite these warnings, random-split evaluation remains prevalent in smartphone- and MEMS-based machine diagnosis, including the baseline accompanying the dataset used here [20]. A rigorous, deployment-honest re-evaluation of low-rate MEMS condition monitoring is therefore still missing.

Table 1. Positioning of this work against representative prior studies. ✓ = addressed, ∼ = partially addressed, ✗ = not addressed.

Representative work(s)	Sensor / rate	Low cost	Leakage-free evaluation	Acc.–resource trade-off	IoT/edge analysis	Sensor charact.
High-rate vibration FD & deep models [5, 6, 7, 4]	Industrial accel., 8–200 kHz	✗	✗	✗	✗	✗
Evaluation / leakage studies [17, 18, 19]	Industrial accel., kHz	✗	✓	✗	✗	✗
Smartphone / mobile machine FD [10, 11, 12]	Smartphone MEMS, $\leq$ kHz	✓	✗	✗	∼	✗
Source-dataset baseline [20]	Smartphone MEMS, 100 Hz	✓	✗	✗	✗	∼
Low-cost MEMS sensor nodes [13, 14]	MEMS node, kHz	✓	✗	∼	✓	∼
TinyML edge fault diagnosis [22, 23]	(Lab) accel., kHz	✓	✗	∼	✓	✗
This work	**Smartphone-grade MEMS, 100 Hz**	✓	✓	✓	✓	✓

2.5 Positioning and Contributions

Table 1 situates this work against representative prior art. Existing studies advance one axis at a time. High-rate benchmarks and deep models maximise accuracy but ignore cost and evaluation rigour; methodology papers expose leakage but on industrial-grade data; smartphone-based studies embrace low-cost sensing but retain optimistic random-split evaluation; and IoT/TinyML works demonstrate edge deployment yet assume industrial bandwidth. To the best of the authors' knowledge, no prior study jointly (i) evaluates ultra-low-rate smartphone-MEMS machine diagnosis under a leakage-free, cross-condition protocol, (ii) characterises the consumer MEMS channel as a measurement instrument, and (iii) maps the accuracy–resource trade-off into an IoT link budget and edge-model footprint. This paper closes that gap, providing a reproducible and deployment-realistic baseline for IoT-based condition monitoring of induction machines with low-cost MEMS sensing.

3 Dataset and Measurement Setup

The present study is based on the publicly available smartphone-acquired vibration dataset introduced in [20] and archived on Mendeley Data [24]. The dataset was selected because it embodies exactly the low-cost, low-bandwidth acquisition regime targeted in this work. This section summarises the test rig, the acquisition chain, the emulated fault conditions, and the organisation of the recordings; the reader is referred to the original data article for the complete experimental protocol. The acquisition and dataset parameters are consolidated in Table 2.

3.1 Test Rig and Machine

The measurements were carried out on a 1.1 kW three-phase squirrel-cage induction machine, a rating representative of the large population of small general-purpose drives found in industry. Variable-speed operation was obtained through a voltage/frequency (V/f)-controlled inverter, and a mechanical load was applied by a DC generator coupled directly to the machine shaft. The generator fed an adjustable resistive load bank, monitored with a digital multimeter, so that the load level could be set to give both loaded and unloaded operation. A consumer smartphone was affixed to the machine terminal box with double-sided adhesive tape, oriented approximately parallel to the shaft axis; the mounting location was marked so that the placement could be reproduced across successive installations. The complete arrangement—induction machine, V/f inverter, coupled DC generator, resistive load bank, and supply-monitoring instrumentation—is shown in Fig. 1.

3.2 Acquisition Chain and Signal Channels

Vibration signals were acquired with an iPhone 15 Pro Max using its integrated tri-axial MEMS inertial sensor, accessed through the *Sensor Play* application. Six channels were recorded for every operating condition: the three raw acceleration components (g_x, g_y, g_z) and the three gravity-compensated linear-acceleration components (g_{userx}, g_{usery}, g_{userz}) produced by the device's on-board sensor fusion. All channels were sampled at 100 Hz, which corresponds to the maximum stable rate supported by the acquisition application, and each operating condition was recorded continuously for 15 minutes. The resulting sampling rate is one to three orders of magnitude below that of the industrial-grade benchmarks discussed in Section 2, and constrains the analysable band to below 50 Hz in accordance with the Nyquist criterion [15, 16].

3.3 Health States and Fault Emulation

Six machine health states were considered: one healthy (fault-free) baseline and five fault conditions spanning mechanical and electrical origins, as summarised in Table 3. The bearing faults were introduced by substituting a 6205ZZ bearing that had been extracted from industrially operated motors, so that the degradation reflects realistic in-service mechanisms rather than artificial machining. Two lubrication-starvation severities were distinguished: insufficient lubrication (B1) and severe insufficient lubrication (B2), the severity being assigned qualitatively from the rotational behaviour observed during manual inspection; a bearing with a cracked outer ring (B3) constituted a localised mechanical defect. The electrical fault, voltage imbalance (V), was produced by inserting voltage-divider resistors in one supply phase, which yielded a pronounced phase-current asymmetry (for example, phase currents of 0.75/2.75/3.03 A at 50 Hz under load [20]). The rotor fault, broken rotor bar (R), was emulated by drilling four holes of 5–9 mm diameter and 16–18 mm depth into the rotor bars. Because fault-related vibration signatures are speed-dependent, the shaft speed was recorded for every condition; the measured values are reproduced in Table 4, where the reduced speed under voltage imbalance (e.g., 1326 r/min versus 1434 r/min for the healthy machine at 50 Hz under load) is consistent with the torque disturbance introduced by that fault.

3.4 Operating Conditions and Dataset Organisation

Each of the six health states was recorded at three supply frequencies (30, 40, and 50 Hz) and under two load conditions (loaded and unloaded), producing a fully crossed matrix of $6 \times 3 \times 2 = 36$ recordings. The files follow the naming convention `Class_Load_SpeedHz`, where the load indicator is `1` for loaded and `0` for unloaded operation; for instance, `B1_1_40Hz` denotes insufficient lubrication under load at 40 Hz. Each recording is provided in both comma-separated (`.csv`) and MATLAB (`.mat`) formats. Representative gravity-compensated waveforms for the six health states, recorded under identical operating conditions, are shown in Fig. 2; the healthy machine exhibits markedly lower vibration amplitude, whereas the fault conditions produce larger and morphologically distinct responses.

3.5 Data Integrity

Prior to analysis, an independent integrity check was performed on all 36 recordings. Each file was confirmed to contain exactly 89,999 samples per channel, corresponding to 900 s at 100 Hz, with no missing values and no systematic sample-and-hold duplication (the maximum fraction of consecutive duplicated rows in any recording was below 0.01%). The channel set and recording length were therefore uniform across the entire dataset, and no imputation or resampling was required.

4 Sensor and Signal Characterisation

Before any classifier is trained, it is instructive to examine the smartphone accelerometer as the front end of the measurement chain and to establish what fault-related information its channels can, in principle, convey. The full metrological characterisation of MEMS accelerometers—calibration, traceability, and frequency response against a reference—has itself become an active subject in the measurement community [27, 28, 25]; such bench measurements are outside the scope of this dataset study, so the characterisation here is deliberately *data-derived*: the effective resolution, a broadband signal floor, the usable frequency band imposed by the 100 Hz sampling rate, and the relative diagnostic value of the raw and gravity-compensated channels are all estimated directly from the acquired recordings, without bench calibration or model training. These estimates delimit what the channel can convey and motivate the sensing choices analysed later.

4.1 Effective Resolution and Noise Floor

The effective amplitude resolution was estimated as the smallest positive difference between consecutive distinct sample values in a healthy, unloaded recording. A quantisation step of approximately 0.015 mg was obtained for the raw acceleration channels and 0.001 mg for the

Table 2. Acquisition and dataset specifications.

Parameter	**Value**
Machine	1.1 kW three-phase squirrel-cage IM
Speed control	V/f-controlled inverter
Loading	DC generator (shaft-coupled)
Sensor	iPhone 15 Pro Max, tri-axial MEMS
Acquisition app	Sensor Play
Channels	6 (g_x, g_y, g_z; $g_{userx}, g_{usery}, g_{userz}$)
Sampling rate	100 Hz
Record length	15 min (89,999 samples/channel)
Health states	6 (H, B1, B2, B3, V, R)
Supply frequencies	30, 40, 50 Hz
Load conditions	loaded, unloaded
Recordings	36 ($6 \times 3 \times 2$)
Formats	`.csv`, `.mat`
Repository	Mendeley Data, CC BY 4.0

Table 3. Health states and fault-emulation summary.

Code	**Health state**	**Emulation**
H	Healthy	Fault-free baseline
B1	Insufficient lubrication	Reduced lubricant, 6205ZZ bearing; severity set by manual inspection
B2	Severe insufficient lubrication	Advanced lubricant starvation; severity set by manual inspection
B3	Cracked outer ring	Bearing with cracked outer ring (ex-service degradation)
V	Voltage imbalance	Voltage-divider resistors in one supply phase; phase-current asymmetry
R	Broken rotor bar	Four holes drilled in rotor bars (5–9 mm diameter, 16–18 mm depth)

gravity-compensated channels, the finer step of the latter reflecting the additional numerical processing of the on-board sensor fusion. A broadband signal floor was estimated from the power spectral density (PSD) of a healthy recording; the median PSD above 30 Hz was 8.2×10^{-8} g^2/Hz (-70.8 dB re 1 g^2/Hz). Because this value is obtained on a *running* machine, it includes residual mechanical vibration and therefore upper-bounds, rather than isolates, the sensor's intrinsic noise floor; a stationary-sensor measurement was not available for this dataset. Even as an upper bound, however, it lies several orders of magnitude below the fault-induced vibration, which reaches amplitudes on the order of 0.1–0.2 g (Fig. 2); sensor resolution and noise are thus not the limiting factor for the coarse fault-discrimination task considered here.

4.2 Usable Frequency Band

At a sampling rate of 100 Hz the analysable spectrum is bounded by the Nyquist frequency of 50 Hz [15, 16]. Fig. 3 shows the Welch PSD of the gravity-compensated g_{userx} channel for the six health states under identical operating conditions. Two observations follow. First, discriminative structure is present within the 0–50 Hz band: the healthy machine exhibits a spectrum roughly an order of magnitude lower in power than the fault conditions, and distinct spectral peaks appear for the rotor and electrical faults. The dominant peaks near 24 Hz coincide with the shaft rotational frequency at 50 Hz supply (Table 4, e.g. 1425–1486 r/min $\approx$ 23.8–24.8 Hz), confirming that the low-frequency band captures the rotational fundamental and its low-order harmonics. Second, the characteristic bearing defect frequencies (ball-pass and ball-spin frequencies and their harmonics) lie largely above 50 Hz at the higher supply frequencies—although some low-order components fall within the band at 30 Hz supply—and are, for the most part, not represented. The dataset is therefore better suited to time-domain and low-frequency analysis than to the high-frequency

Table 4. Measured shaft speed (r/min) for each health state, supply frequency, and load condition (reproduced from [20]).

Supply	Load	H	B1	B2	B3	V	R
30 Hz	Loaded	833.9	829.4	833.0	833.8	787.0	815.8
	Unloaded	892.7	888.3	892.7	896.3	893.0	894.9
40 Hz	Loaded	1136	1138	1138	1140	1076	1122
	Unloaded	1190	1192	1193	1192	1192	1192
50 Hz	Loaded	1434	1436	1439	1436	1326	1425
	Unloaded	1486	1490	1491	1489	1486	1486

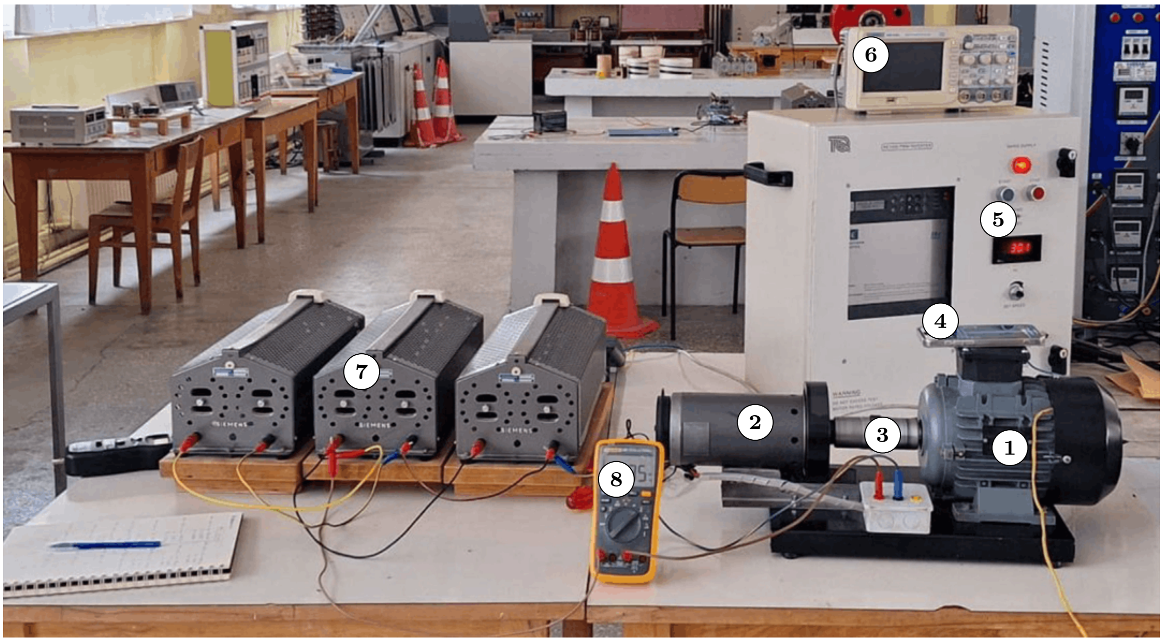

Figure 1. Test-rig arrangement (photograph adapted from [20]): (1) squirrel-cage induction machine under test; (2) shaft-coupled DC generator providing the mechanical load; (3) flexible shaft coupling; (4) smartphone supplying the tri-axial MEMS vibration signal; (5) V/f variable-frequency supply; (6) oscilloscope; (7) resistive load bank; (8) digital multimeter.

envelope analysis conventionally used for incipient bearing diagnosis, and a corresponding difficulty in separating the fine lubrication-severity states is anticipated.

4.3 Raw versus Gravity-Compensated Channels

The dataset provides both the raw acceleration and the gravity-compensated (*userAcceleration*) triads, and their relative diagnostic value is not self-evident. Two questions were separated: whether the on-device gravity compensation removes vibration *power*, and whether it removes diagnostic *information*. For the first, the alternating-current (AC) power of each raw channel was compared with that of its gravity-compensated counterpart over every recording; the mean raw-to-compensated power ratio was 1.00 on all three axes (0.999, 1.001, 1.000 for x, y, and z). The compensation therefore does not attenuate the vibration power. It does, however, *reshape* it: the mean-removed raw signal and the compensated signal are only moderately correlated (Pearson 0.58, 0.46, and 0.46 for x, y, and z), so the complementary filter redistributes the low-frequency content—altering its phase and correlation structure—rather than discarding it, and the two triads are complementary rather than redundant.

For the second question, the mutual information between each channel's time-domain feature set and the fault label was estimated (Fig. 4). The raw channels were consistently more informative (mean mutual information 0.69) than the gravity-compensated channels (0.50), a ratio of 1.39. Because the raw channels also retain a per-recording static offset that varies with condition, a legitimate concern is that this advantage merely encodes recording identity—the very leakage this work otherwise criticises. Two controls indicate that it does not: removing the offset-dependent *mean* feature leaves the ratio essentially unchanged (1.31), and restricting the estimate to a DC-invariant feature subset (standard deviation, variance, skewness, kurtosis, zero-crossing rate,

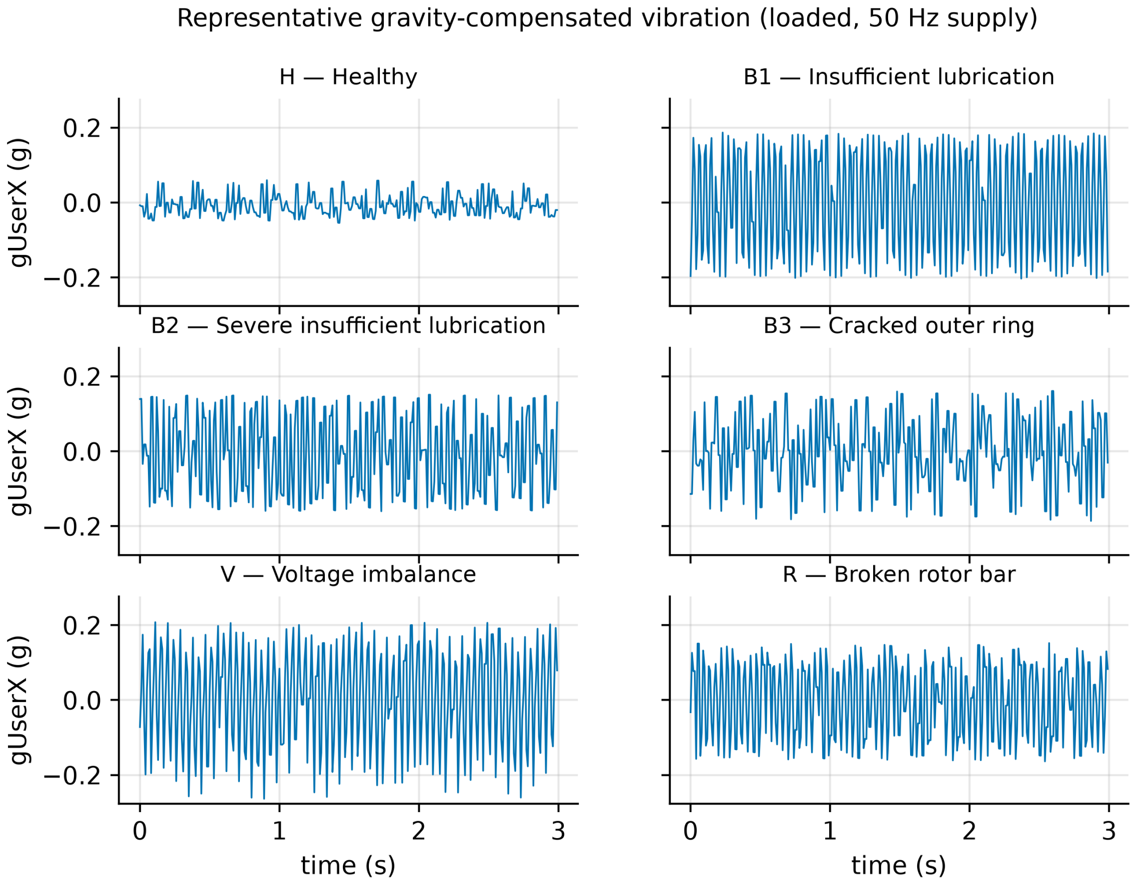

Figure 2. Representative gravity-compensated vibration (g_{userx}, 3 s excerpt) for the six health states under identical operating conditions (loaded, 50 Hz supply). Amplitudes share a common scale.

and amplitude entropy) yields a comparable ratio (1.32). The raw-channel advantage is thus robust to the removal of static offsets and reflects genuine dynamic content. The raw channels are therefore preferable for condition monitoring, and this informativeness ordering motivates the channel-selection study reported in Section 6.

5 Methodology

The processing chain evaluated in this work is deliberately lightweight, so that it is representative of what an inexpensive sensor node can execute: fixed-length windowing, a compact set of time-domain features, and a conventional classifier. The chain is illustrated in Fig. 5. The emphasis of this section, and the principal methodological contribution of the paper, is the set of *evaluation protocols* used to obtain deployment-realistic performance estimates.

5.1 Segmentation and Windowing

Let the dataset comprise $R = 36$ continuous recordings, each associated with a health label and an operating condition (supply frequency and load). Every recording was divided into fixed-length windows of 2 s with 50% overlap, matching the segmentation of the source data article. At 100 Hz this yields windows of 200 samples and a total of 32,328 windows, distributed equally across the six classes (5,388 per class). The window length and overlap were treated as configurable parameters and were varied in the trade-off study of Section 6.

5.2 Feature Extraction

Each window was reduced to a set of 14 time-domain features per channel, summarised in Table 6. These comprise standard statistical and shape descriptors widely used in vibration-based diagnosis, and require no spectral transform, which keeps the extraction cost compatible with a microcontroller. For a window of C channels, the per-channel feature vectors were concatenated into a single descriptor of length $14C$ (for example, 84 for the full six-channel set). The six-feature subset marked in Table 6, corresponding to the descriptors used by the source data article's

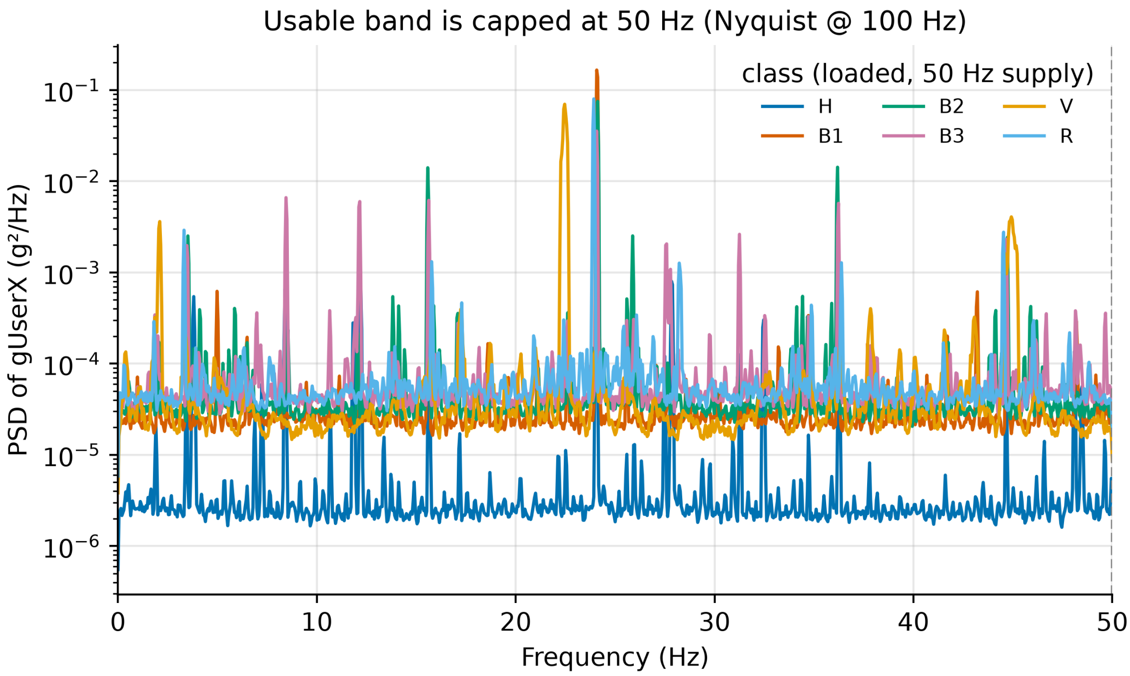

Figure 3. Welch power spectral density of the g_{userx} channel for the six health states (loaded, 50 Hz supply). The analysable band is bounded by the 50 Hz Nyquist frequency; the healthy state is markedly lower in power, and rotor/electrical faults produce distinct low-frequency peaks.

baseline, was retained for a feature-set ablation. For the distance- and margin-based classifiers, features were standardised to zero mean and unit variance using statistics estimated on the training partition only.

5.3 Classifiers

Three baseline classifiers were considered, chosen for their prevalence in the fault-diagnosis literature and their modest computational footprint: a Random Forest (RF) with 200 trees; a k-nearest-neighbour classifier (KNN, $k = 5$); and a support vector machine (SVM) with a radial-basis-function kernel. To keep the SVM tractable on the full window set, the kernel was approximated by a Nyström feature map (400 components) followed by a linear SVM. For the edge-footprint analysis of Section 6, two additional lightweight models—a depth-limited decision tree and a multinomial logistic regression—were included. All models map a feature descriptor to one of the six health states.

5.4 Evaluation Protocols

The central methodological question is how the labelled windows are partitioned into training and test sets. Four protocols were defined and applied to identical features so that differences in reported accuracy are attributable solely to the partitioning strategy.

5.4.1 RANDOM A stratified random split over all windows, as commonly used in the literature and by the source baseline. This protocol is retained as a reference precisely because it is susceptible to leakage: adjacent windows share 50% of their samples, and all windows of a recording share the same operating point and sensor coupling, so a model may recognise the originating recording rather than the fault.

5.4.2 GROUP A leakage-free protocol in which entire recordings are held out. Stratified group k-fold cross-validation ($k = 6$) was used, with the recording identifier as the grouping variable, so that no window from a test recording appears in training. With six recordings per class, each fold holds out one recording per class.

5.4.3 CROSS-SPEED A leave-one-supply-frequency-out protocol: the model is trained on two supply frequencies and tested on the third, and results are averaged over the three folds. This quantifies generalisation to an unseen operating speed.

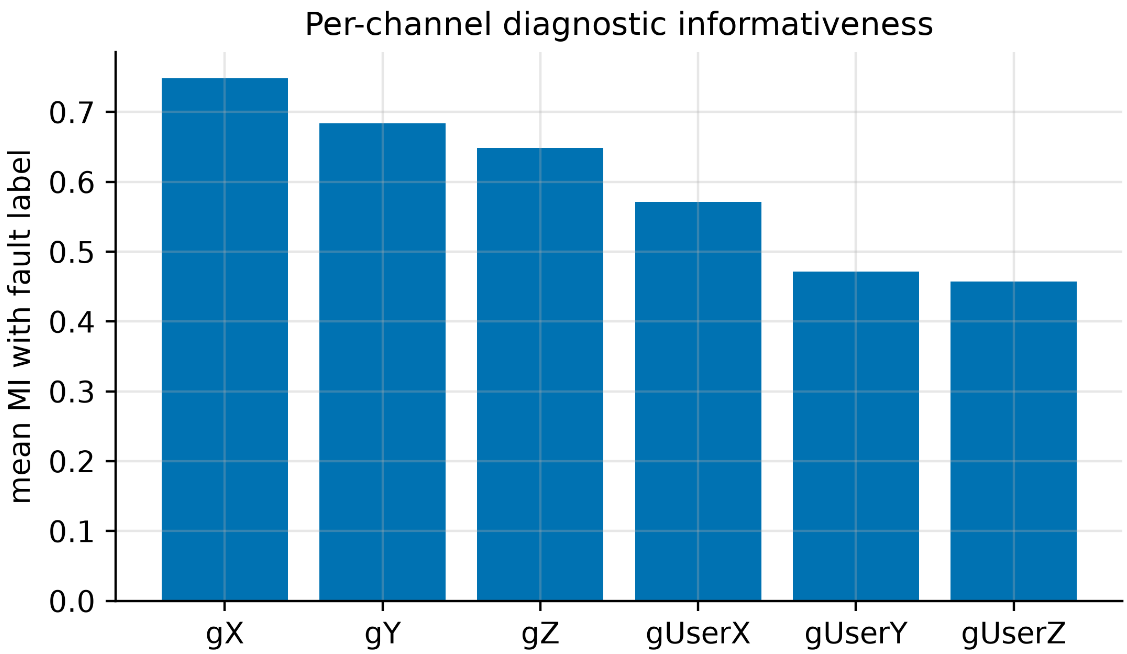

Figure 4. Mean mutual information between each channel's time-domain feature set and the fault label. The raw channels (g_x, g_y, g_z) are more informative than the gravity-compensated channels (g_{userx}, g_{usery}, g_{userz}).

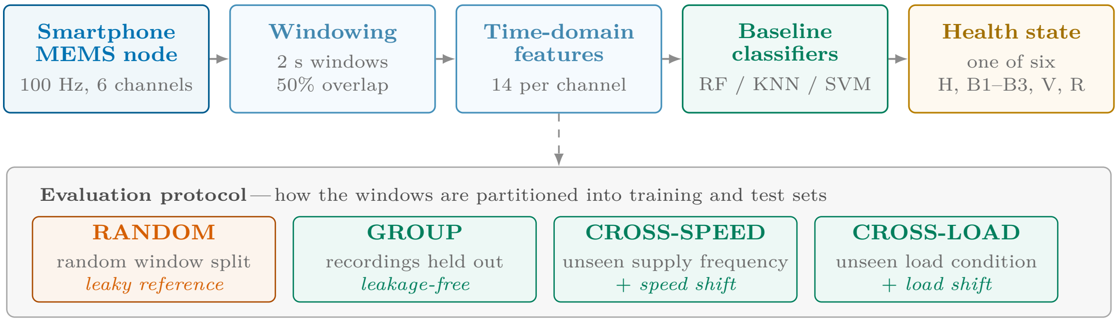

Figure 5. Processing chain and evaluation protocols. Identical features are assessed under four data-partitioning protocols: RANDOM is retained only as a leaky reference, whereas GROUP, CROSS-SPEED and CROSS-LOAD are leakage-free at the recording level, the latter two additionally imposing an operating-condition shift. The protocol, not the model, is the primary object of study.

5.4.4 CROSS-LOAD A leave-one-load-out protocol: the model is trained on one load condition (loaded or unloaded) and tested on the other. This quantifies generalisation to an unseen load.

The GROUP, CROSS-SPEED, and CROSS-LOAD partitions are all group-disjoint at the recording level and therefore free of the window-overlap leakage that affects the RANDOM protocol; the latter two additionally introduce a deliberate operating -condition shift between training and test.

5.5 Statistical Evaluation and Uncertainty

Classification performance was reported as accuracy and macro-averaged F_1-score, the latter being appropriate for the balanced six-class problem while remaining sensitive to per-class failures. To account for the limited number of recordings, the group-aware protocols were repeated over five random seeds, each of which reshuffles the recording-level fold composition, so that the five runs constitute five independent re-partitionings of the same data. The mean accuracy across the five seeds is reported together with its Type A expanded uncertainty [29], obtained by treating the five per-seed mean accuracies as repeated measurements and applying a coverage factor $k = t_{0.975,\,4} \approx 2.78$. For the headline configuration comparisons, the same five seed-defined partitions were applied to every configuration, so that each configuration pair is evaluated on identical held-out recordings; significance was then assessed on the five per-seed paired differences with a paired t-test and a Wilcoxon signed-rank test. The seed—an independent re-partitioning—is

Table 5. Estimator and feature-extraction settings. Parameters not listed were left at their library defaults; d denotes the feature-vector dimension ($14C$ for C channels).

Component	Setting
Random Forest (baseline)	200 trees, unlimited depth, Gini split, $\sqrt{d}$ features per split
Random Forest (edge)	50 trees, maximum depth 8
Decision tree (edge)	maximum depth 8
KNN	$k = 5$, Euclidean metric, uniform weights
SVM	RBF kernel approximated by a Nyström map (400 components, $\gamma = 1/d$), then a linear SVM ($C = 10$, squared hinge, primal solver, ≤ 5000 iterations)
Logistic regression	multinomial, L_2 penalty, $C = 1$, L-BFGS, ≤ 2000 iterations
Standardisation	zero mean, unit variance; fitted on the training partition only (KNN, SVM, logistic regression)
Amplitude entropy	32-bin normalised histogram per window
Bit-depth reduction	per-recording, per-channel min–max range uniformly quantised to $2^b - 1$ levels
Packet loss	independent Bernoulli per sample, concealed by sample-and-hold
Orientation change	Rodrigues rotation of both triads by 30° about the axis $(1, 1, 0)/\sqrt{2}$

thus the experimental unit ($n = 5$); pooling the 5×6 seed-by-fold accuracies as if independent would pseudoreplicate and overstate significance, and is avoided. This expanded uncertainty reflects only the reproducibility of the evaluation on the available data—the seed-to-seed re-partitioning variance—and does not capture the additional, unquantified uncertainty arising from the single-machine, single-specimen scope (Section 7.2); it should therefore be read as a reproducibility interval rather than a full measurement-uncertainty budget. Confusion matrices were aggregated from the pooled out-of-fold predictions of the GROUP protocol. All experiments used the fixed seeds 0–4, and the feature set (Table 6), estimator settings (Table 5), and partitioning rules are reported in full, so that the results can be reimplemented independently.

5.6 Implementation Details

All estimators were taken from a single, standard machine-learning library, with the settings collected in Table 5; any parameter not listed was left at its library default, and every stochastic component was seeded. Three further choices affect reproducibility and are therefore stated explicitly. First, decimation to a lower sampling rate was performed by polyphase resampling with the associated anti-alias FIR filter, so that the reduced-rate conditions of Section 6.2 combine a change of rate with the low-pass that necessarily accompanies it. Second, the mutual information of Section 4 was obtained with a k-nearest-neighbour estimator ($k = 3$) applied to the continuous features and reported in nats; the estimator is non-negative by construction, so values near zero indicate no detectable dependence rather than independence. Third, the power spectral densities were computed by Welch's method with a Hann window, 50% segment overlap and density scaling, using 2048-sample segments for the per-class spectra of Fig. 3 and 1024-sample segments for the signal-floor estimate.

6 Results

6.1 The Optimism Gap of Random-Split Evaluation

Table 7 reports the classification accuracy of the three baseline models under the four evaluation protocols, and Fig. 6 visualises the same result. Under the RANDOM protocol every model attains essentially perfect accuracy, reproducing—and slightly exceeding—the 98% reported by the source data article's random-split baseline. Under the leakage-free GROUP protocol the accuracy of the same features and models falls to roughly one-half, an *optimism gap* of approximately 41 to 47 accuracy points. The CROSS-SPEED and CROSS-LOAD protocols, which additionally impose an operating-condition shift, yield comparably reduced accuracy. The mechanism is the expected one: because overlapping windows share half of their samples and every window of a recording shares one operating point and sensor coupling, a randomly partitioned model can identify the originating recording rather than the fault. Two quantitative observations follow from Table 7: the leakage-free accuracy remains well above the 16.7% chance level, and the lower-capacity models (KNN, SVM) attain higher GROUP accuracy than the Random Forest.

The seed-averaged confusion matrix of the GROUP protocol (Fig. 7) localises the residual difficulty. The healthy state is the most reliably identified, followed by the supply-imbalance and broken-rotor-bar faults, whereas the two lubrication-severity states (B1 and B2) are frequently

Table 6. Per-channel time-domain features. For a window $x = \{x_i\}_{i=1}^{W}$ with mean μ and standard deviation σ. Features marked † form the six-feature ablation subset.

Feature	**Definition**
Mean	$\mu = \frac{1}{W}\sum_i x_i$
Std. dev.†	$\sigma = \sqrt{\frac{1}{W}\sum_i (x_i - \mu)^2}$
Variance	σ^2
RMS†	$x_{\mathrm{rms}} = \sqrt{\frac{1}{W}\sum_i x_i^2}$
Peak-to-peak†	$\max_i x_i - \min_i x_i$
Max. absolute	$x_{\max} = \max_i \lvert x_i \rvert$
Skewness†	$\frac{1}{W}\sum_i (x_i - \mu)^3/\sigma^3$
Kurtosis†	$\frac{1}{W}\sum_i (x_i - \mu)^4/\sigma^4 - 3$
Crest factor†	$x_{\max}/x_{\mathrm{rms}}$
Shape factor	$x_{\mathrm{rms}}/\big(\frac{1}{W}\sum_i \lvert x_i \rvert\big)$
Impulse factor	$x_{\max}/\big(\frac{1}{W}\sum_i \lvert x_i \rvert\big)$
Clearance factor	$x_{\max}/\big(\frac{1}{W}\sum_i \sqrt{\lvert x_i \rvert}\big)^2$
Zero-crossing rate	rate of sign changes of $x_i - \mu$
Amplitude entropy	$-\sum_k p_k \log_2 p_k$ (32-bin histogram)

confused with each other and with the remaining bearing conditions, in agreement with the 50 Hz band limit established in Section 4. Because this per-class behaviour is markedly uneven, accuracy alone overstates how well the fine bearing states are resolved: the macro-averaged F_1-score of the same configuration is 51.1%, and per-class recall falls from 0.87 for the healthy state and 0.66 for the supply imbalance to 0.35 for B3 and 0.16 for B2. The lubrication-severity states therefore dominate the residual error, and the headline figures below should be read with that qualification.

6.2 Accuracy–Resource Trade-offs

Each sensing and communication parameter was varied in turn under the GROUP protocol (Fig. 8). Reducing the sampling rate did not degrade accuracy: as shown in Table 8, every rate at or below 25 Hz matched or exceeded the accuracy obtained at 100 Hz, with a maximum near 25 Hz. Reducing the channel set was likewise not detrimental—the three raw axes alone outperformed the full six-channel set—while lengthening the window from 2 s to 4 s increased accuracy at the cost of detection latency.

The combination of these frugal choices was then examined (Table 9 and Fig. 9). The resource-optimal configuration—25 Hz sampling, the three raw channels, and a 4 s window—raised the Random Forest accuracy from 52.8% at the naive 100 Hz/six-channel/2 s baseline to 75.9%, while reducing the raw data volume by approximately eight-fold. Robustness to further deployment stresses was also measured: reducing the amplitude resolution to 4 bits lowered accuracy by only a few points, injected packet loss of up to 20% left accuracy essentially unchanged, and the six-feature subset of the source baseline performed within noise of the full feature set. The largest sensitivity was to sensor mounting: a simulated 30° change in orientation degraded accuracy substantially.

To establish that these differences exceed the measurement uncertainty, the baseline, the 25 Hz variant, and the resource-optimal configuration were compared across five random recording-level partitions, one per seed, so that within each partition every configuration is evaluated on identical held-out recordings and the five per-partition differences form a paired sample. Table 10 reports the resulting accuracies with their Type A expanded uncertainty (coverage factor $k \approx 2.78$). The paired comparisons confirm the trade-off findings: the resource-optimal configuration classifies the health state 23.1 percentage points more accurately than the baseline (paired t-test $p = 0.0001$), and reducing the sampling rate from 100 to 25 Hz alone accounts for 10.1 of those points ($p = 0.003$). These tests use the seed—an independent re-partitioning of the data—as the experimental unit ($n = 5$), rather than the 5×6 seed-by-fold pseudoreplicates that would spuriously inflate significance; at this honest unit the Wilcoxon signed-rank test attains its smallest attainable value for $n = 5$ ($p = 0.0625$) and is reported only for completeness, whereas the parametric test, which retains power at this sample size, supports the improvements. Consistent with this, the expanded-uncertainty intervals of the three configurations in Table 10 do not overlap.

Table 7. Accuracy (%) by evaluation protocol and model (fixed recording-level partition, five seeds; mean ± Type A expanded uncertainty, $k \approx 2.78$; 100 Hz, six channels, 2 s).

Protocol	**RF**	**KNN**	**SVM**
RANDOM (leaky)	100.0	100.0	100.0
GROUP (recording-holdout)	52.8 ± 4.4	59.0 ± 1.4	58.4 ± 3.6
CROSS-SPEED	49.9 ± 1.3	48.2 ± 0.0	50.1 ± 1.0
CROSS-LOAD	49.5 ± 2.7	40.3 ± 0.0	40.1 ± 0.5

Table 8. Sampling-rate sweep under the GROUP protocol (Random Forest; fixed recording-level partition, five seeds; mean ± Type A expanded uncertainty, $k \approx 2.78$). Lower rates match or exceed 100 Hz.

Rate (Hz)	**Raw bitrate (kbps)**	**Accuracy (%)**
100	9.6	52.8 ± 4.4
50	4.8	39.2 ± 3.5
25	2.4	$\mathbf{62.9 \pm 2.5}$
20	1.9	60.9 ± 6.0
12.5	1.2	60.9 ± 3.2
10	1.0	58.8 ± 0.9

6.3 Deployment Implications: Communication and Edge Footprint

Table 11 compares the bitrate of three telemetry strategies with the sustainable throughput of common low-power radios. Streaming the raw six-channel signal requires 9.6 kbps and transmitting the extracted feature vector requires 2.69 kbps; both are carried by Bluetooth Low Energy, Zigbee, or NB-IoT, but neither fits within the duty-cycle-limited throughput of a LoRaWAN uplink, which admits only the 40 bps on-node decision stream. Table 12 lists the leakage-free accuracy and serialised size of four candidate models: a 4.7 kB logistic-regression classifier exceeded the accuracy of a 1.09 MB Random Forest, and the time-domain feature extraction requires only $\mathcal{O}(WC)$ operations per window with no spectral transform.

7 Discussion and Limitations

7.1 Discussion

7.1.1 Evaluation methodology The optimism gap of Section 6.1 is the central methodological finding: for low-rate smartphone-based machine diagnosis, accuracies obtained under random splitting of overlapping windows overstate deployable performance by tens of accuracy points. The effect mirrors that documented for industrial-grade vibration data [18, 19] and the broader machine-learning leakage literature [17], but its magnitude here is large because each operating condition is represented by a single continuous recording. A practical consequence is that the near-perfect accuracies frequently reported for this class of data should be interpreted with caution; the recording-wise and cross-condition figures obtained here provide a more deployment-honest reference against which future methods can be compared.

7.1.2 Low sampling rate is not a barrier Decimation to 25 Hz did not reduce accuracy and in several runs improved it, so the low sampling rate of consumer MEMS sensors is not a barrier to coarse-fault discrimination. This observation warrants care rather than celebration: the sweep is non-monotonic—the intermediate 50 Hz rate performed worst—and, because decimation applies an anti-alias low-pass, the gain at 25 Hz is entangled with the removal of high-frequency content that the classifier would otherwise overfit on the small training set, rather than with the sample rate as such. The defensible conclusion is the weaker but still useful one that the discriminative information for these faults resides in the low-frequency band identified in Section 4, so that aggressive decimation reduces telemetry and, by extension, node energy without sacrificing accuracy. On this dataset the resource-frugal configuration established in Section 6.2 —25 Hz sampling, the three raw channels, and a 4 s window—is a strong operating point; confirming it as a general default would require the multi-machine validation identified in Section 7.2.

7.1.3 Model complexity and edge feasibility Under leakage-free evaluation the lower-capacity models generalised at least as well as the Random Forest while occupying a kilobyte-scale footprint, so on-node inference is practical on inexpensive microcontrollers. Combined with the link budget,

Table 9. Incremental frugal recipe under the GROUP protocol (fixed recording-level partition, five seeds; mean $\pm$ Type A expanded uncertainty, $k \approx 2.78$). The resource-optimal node exceeds the naive baseline.

Configuration	RF	KNN	LogReg
Baseline (100 Hz, 6 ch, 2 s)	52.8 ± 4.4	59.0 ± 1.4	59.6 ± 2.8
+ 25 Hz	62.9 ± 2.5	46.5 ± 2.0	57.3 ± 1.7
+ raw 3 ch	65.6 ± 1.8	51.4 ± 1.7	52.5 ± 2.3
+ 4 s window (optimal)	$\mathbf{75.9 \pm 2.8}$	54.9 ± 2.7	60.3 ± 3.9

Table 10. Measured leakage-free (GROUP) accuracy of the Random Forest with Type A expanded uncertainty ($k \approx 2.78$, from five random recording-level partitions), and paired comparisons; within each partition all configurations are evaluated on identical held-out recordings, and significance is assessed on the five per-partition paired differences.

Configuration	Accuracy (%)
Baseline (100 Hz, 6 ch, 2 s)	52.8 ± 4.4
25 Hz, 6 ch, 2 s	62.9 ± 2.5
Optimal (25 Hz, 3 raw ch, 4 s)	$\mathbf{75.9 \pm 2.8}$
Paired comparison (Δ, paired t-test, $n = 5$)	
Optimal − Baseline	$+23.1$, $p = 0.0001$
25 Hz − 100 Hz	$+10.1$, $p = 0.003$

this yields per-link deployment guidance: Bluetooth Low Energy, Zigbee, and NB-IoT nodes can stream raw or feature data to a gateway, whereas LoRaWAN nodes, constrained by the duty cycle to the decision stream, must perform classification locally. The alignment between the models that deploy most easily and those that generalise best is convenient for practical systems.

7.1.4 Sensor-level guidance The channel analysis of Section 4 carries a concrete recommendation for smartphone-based acquisition: the raw acceleration channels should be preferred over the gravity-compensated channels. The on-device gravity compensation—valuable for motion-tracking applications—preserves the vibration power but reshapes its low-frequency structure, and the raw channels carry measurably more fault information (a mutual-information ratio of about 1.4 that survives the removal of static offsets), so they are the better choice for condition monitoring.

7.1.5 Scope of applicability Taken together, the results delineate where low-cost, low-rate MEMS sensing is appropriate. Coarse health states, such as a healthy machine, a broken rotor bar, or a supply-voltage imbalance, are separable at 100 Hz and below; the staging of incipient bearing degradation, whose signatures lie in the high-frequency band, is not, and remains the province of higher-rate, industrial-grade instrumentation.

7.2 Limitations

Several limitations bound the scope of these findings. The dataset comprises a single 1.1 kW machine with laboratory-induced, discrete-severity faults, and the two lubrication states were graded qualitatively; the reported accuracies are therefore a deployment-honest baseline rather than a claim of field performance. Crucially, each fault class is represented by a *single physical specimen* (one modified bearing, one drilled rotor, one imbalance configuration) recorded across speeds and loads. The recording-wise protocol therefore removes window-overlap and exact operating-point leakage, but not specimen identity: a model may still key on the idiosyncratic signature of that particular specimen and its mounting rather than on a fault signature that would transfer to a different specimen of the same class. The evaluation is thus leakage-free only in this restricted sense, and cross-specimen generalisation is not established here. The small number of recordings (36) also produces wide fold-to-fold variability; although five-seed repetition stabilises the central estimates and the reported expanded uncertainty quantifies their reproducibility, it captures only resampling variance and not the dominant specimen-, machine-, and mounting-to-mounting components, so it should not be read as a full measurement-uncertainty budget. The deployment quantities are likewise indicative rather than measured: the model footprint is a serialised (software) size, and energy and latency are inferred from data volume and window length rather than measured on hardware, with a cycle-accurate microcontroller

Table 11. Communication link budget: telemetry bitrate versus sustainable uplink capacity. Bitrates assume a 1 s reporting hop, `int16` raw samples and `float32` features; the assumed sustainable application-level uplinks are 300 kbps (Bluetooth Low Energy 4.2), 100 kbps (Zigbee, IEEE 802.15.4), 30 kbps (NB-IoT) and 55 bps (LoRaWAN, a spreading-factor-7 physical rate of 5.47 kbps throttled by a 1% duty cycle). ✓ = sustainable, ✗ = infeasible.

Telemetry stream	Bitrate	BLE	Zigbee	NB-IoT	LoRaWAN
Raw (6 ch, int16)	9.6 kbps	✓	✓	✓	✗
Features (84/s)	2.69 kbps	✓	✓	✓	✗
Features (min., 6/s)	192 bps	✓	✓	✓	✗
On-node decision	40 bps	✓	✓	✓	✓

Table 12. Edge-model footprint versus leakage-free (GROUP) accuracy at the baseline configuration (100 Hz, six channels, 2 s; five-partition mean ± Type A expanded uncertainty, $k \approx 2.78$). Serialised sizes are architecture-determined.

Model	GROUP accuracy (%)	Serialised size
Random Forest (200 trees)	52.8 ± 4.4	1.09 MB
Random Forest (50, depth 8)	52.9 ± 4.2	214 kB
Decision tree (depth 8)	44.3 ± 2.4	4.3 kB
Logistic regression	$\mathbf{59.6 \pm 2.8}$	4.7 kB

characterisation left to future work. A further limitation concerns reproducibility in practice: the implementation itself is not distributed, so the results must be reimplemented from the description rather than re-executed. The feature definitions, estimator settings, resampling and robustness transforms, and random seeds are therefore reported in full in Section 5, but exact numerical agreement with an independent reimplementation cannot be guaranteed. Finally, generalisation across machines of different ratings and manufacturers was not assessed and constitutes the most important direction for future validation.

8 Conclusion

This paper examined whether low-cost, low-rate smartphone-MEMS sensing can support condition monitoring of induction machines, and how such monitoring should be evaluated and deployed. Using a public 100 Hz vibration dataset, the near-perfect accuracies reported under random-split evaluation were shown to be largely an artefact of data leakage: under a leakage-free, recording-wise protocol the accuracy of standard classifiers fell by approximately 41 to 47 percentage points, with comparable reductions under cross-speed and cross-load shifts. The residual accuracy nonetheless remained well above chance, confirming that coarse health states—notably a healthy machine and a broken rotor bar—are recoverable at 100 Hz, whereas the fine staging of incipient bearing degradation, whose signatures lie beyond the 50 Hz usable band, is not.

Characterising the consumer sensor as a measurement instrument showed that its on-device gravity compensation preserves the vibration power but reshapes its low-frequency structure, leaving the raw channels the more informative, so that they should be preferred. Most consequentially for deployment, the accuracy–resource analysis established that a resource-frugal configuration—25 Hz sampling, three raw channels, and a 4 s window—not only suffices but outperforms the naive configuration under honest evaluation, raising accuracy to 76% while reducing the raw data volume roughly eight-fold; combined with a communication link budget and a kilobyte-scale edge-model footprint, this yields concrete operating points for Bluetooth Low Energy, Zigbee, NB-IoT, and LoRaWAN nodes. Under honest evaluation, therefore, a smartphone-class sampling rate is not in itself a barrier to IoT-based machine monitoring; on this dataset, moderate decimation to 25 Hz did not reduce—and in fact improved—the measured accuracy, although whether this reflects a genuine anti-aliasing benefit or a dataset-specific effect remains to be established on further machines.

Future work will extend the evaluation across multiple machines of different ratings and manufacturers to establish cross-machine generalisation, and will complement the present analysis with direct, cycle-accurate measurement of on-device inference latency and energy. The evaluation protocol set out in this paper is intended to serve as a deployment-honest baseline for these efforts.

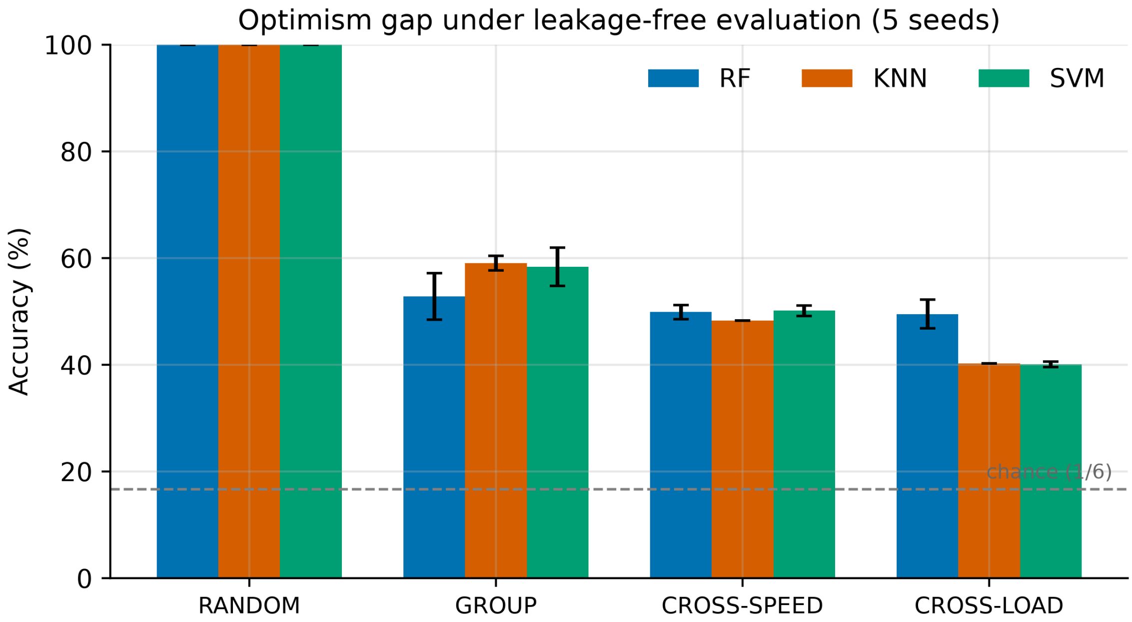

Figure 6. Accuracy by evaluation protocol and model. Near-perfect accuracy under the RANDOM split collapses once evaluation is leakage-free (GROUP) or imposes an operating-condition shift (CROSS-SPEED, CROSS-LOAD).

Data availability statement

The data that support the findings of this study are openly available in Mendeley Data at https://doi.org/10.17632/rs4vz8n3t5.1, under a CC BY 4.0 licence, and are described in the accompanying data article [20, 24]. No new data were created in this study. All processing steps required to reproduce the reported results—the window segmentation, the time-domain feature set (Table 6), the classifier configurations, the partitioning rules of the four evaluation protocols, and the random seeds—are specified in Section 5, including the estimator settings of Table 5.

References

[1] Gangsar P and Tiwari R 2020 *Mechanical Systems and Signal Processing* **144** 106908 ISSN 0888-3270 URL http://dx.doi.org/10.1016/j.ymssp.2020.106908

[2] Hamani K, Kuchar M, Kubatko M and Kirschner S 2025 *Sensors* **25** 5942 ISSN 1424-8220 URL http://dx.doi.org/10.3390/s25195942

[3] Yang Y, Haque M M M, Bai D and Tang W 2021 *Energies* **14** 7017 ISSN 1996-1073 URL http://dx.doi.org/10.3390/en14217017

[4] Zhang S, Zhang S, Wang B and Habetler T G 2020 *IEEE Access* **8** 29857–29881 ISSN 2169-3536 URL http://dx.doi.org/10.1109/ACCESS.2020.2972859

[5] Lee S, Kim T and Kim T 2024 *Data in Brief* **57** 110940 ISSN 2352-3409 URL http://dx.doi.org/10.1016/j.dib.2024.110940

[6] Sehri M and Dumond P 2024 *Data in Brief* **53** 110144 ISSN 2352-3409 URL http://dx.doi.org/10.1016/j.dib.2024.110144

[7] Thuan N D and Hong H S 2023 *BMC Research Notes* **16** ISSN 1756-0500 URL http://dx.doi.org/10.1186/s13104-023-06400-4

[8] Huang H and Baddour N 2018 *Data in Brief* **21** 1745–1749 ISSN 2352-3409 URL http://dx.doi.org/10.1016/j.dib.2018.11.019

[9] Grouios G, Ziagkas E, Loukovitis A, Chatzinikolaou K and Koidou E 2022 *Sensors* **23** 192 ISSN 1424-8220 URL http://dx.doi.org/10.3390/s23010192

[10] Naha A, Thammayyabbabu K R, Samanta A K, Routray A and Deb A K 2017 *IEEE Embedded Systems Letters* **9** 117–120 ISSN 1943-0671 URL http://dx.doi.org/10.1109/LES.2017.2734798

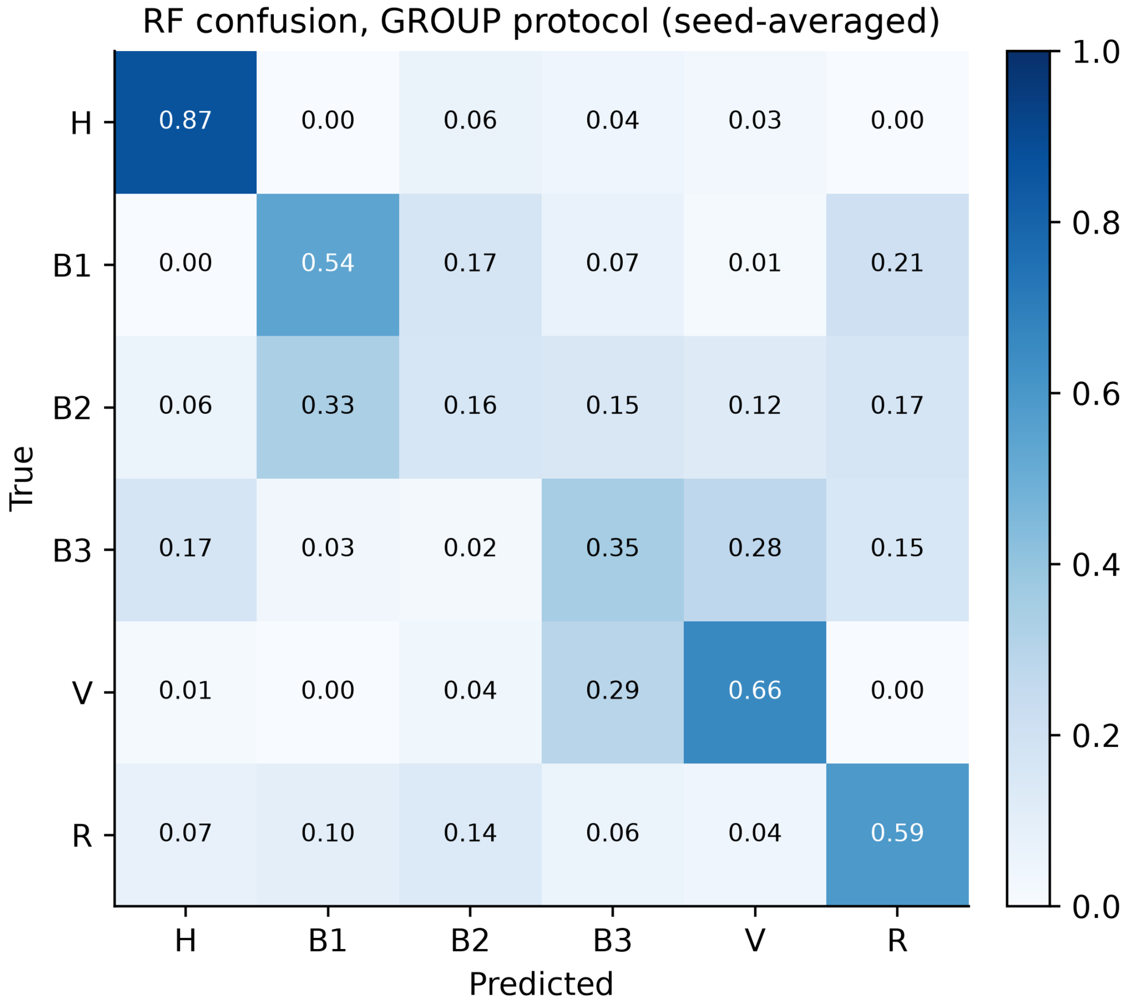

Figure 7. Seed-averaged confusion matrix of the Random Forest under the GROUP protocol (row-normalised; baseline configuration, five recording-level partitions). The healthy state (H) is the most separable, followed by the supply-imbalance (V) and broken-rotor-bar (R) faults; the lubrication-severity states (B1, B2) are the most confused.

[11] Ertarğın M, Gürgenç T, Yıldırım Ö and Orhan A 2023 *European Journal of Technic* ISSN 2536-5010 URL http://dx.doi.org/10.36222/ejt.1336342

[12] Ertarğın M, Orhan A, Yıldırım Ö and Gürgenç T 2025 *Measurement* **253** 117539 ISSN 0263-2241 URL http://dx.doi.org/10.1016/j.measurement.2025.117539

[13] dos Santos Pedotti L A, Zago R M, Giesbrecht M and Fruett F 2020 *IEEE Instrumentation & Measurement Magazine* **23** 25–33 ISSN 1941-0123 URL http://dx.doi.org/10.1109/mim.2020.9234762

[14] Jakobsen M O 2024 *HardwareX* **18** e00525 ISSN 2468-0672 URL http://dx.doi.org/10.1016/j.ohx.2024.e00525

[15] Nyquist H 1924 *Transactions of the American Institute of Electrical Engineers* **XLIII** 412–422 ISSN 2330-9431 URL http://dx.doi.org/10.1109/T-AIEE.1924.5060996

[16] Shannon C E 1948 *Bell System Technical Journal* **27** 379–423 ISSN 0005-8580 URL http://dx.doi.org/10.1002/j.1538-7305.1948.tb01338.x

[17] Kapoor S and Narayanan A 2023 *Patterns* **4** 100804 ISSN 2666-3899 URL http://dx.doi.org/10.1016/j.patter.2023.100804

[18] Wheat L, Mohrenschildt M V, Habibi S and Al-Ani D 2024 *IEEE Access* **12** 169879–169895 ISSN 2169-3536 URL http://dx.doi.org/10.1109/ACCESS.2024.3497716

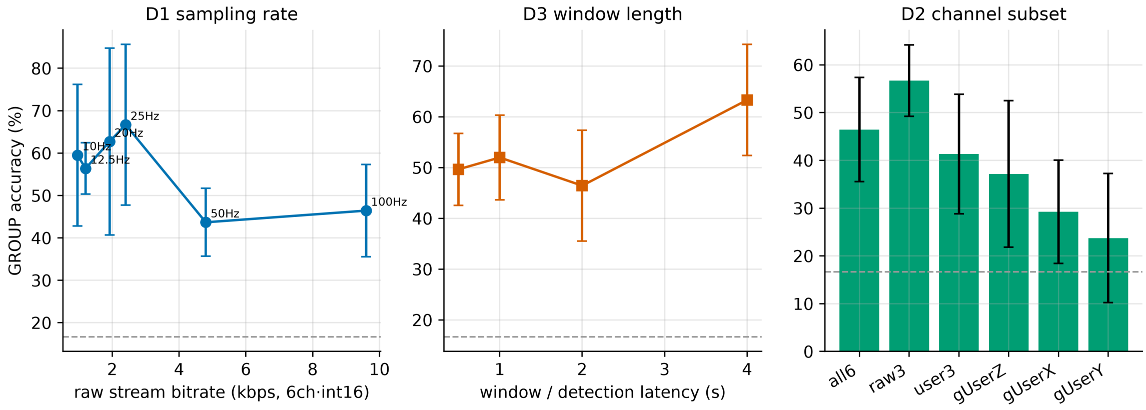

Figure 8. Accuracy–resource trade-offs under the GROUP protocol: sampling rate (left), window length (centre), and channel subset (right). The dashed line marks the 16.7% chance level.

[19] Hendriks J, Dumond P and Knox D 2022 *Mechanical Systems and Signal Processing* **169** 108732 ISSN 0888-3270 URL http://dx.doi.org/10.1016/j.ymssp.2021.108732

[20] Ertarğın M, Orhan A, Yıldırım Ö, Chauhan S and Vashishtha G 2026 *Data in Brief* **67** 112916 ISSN 2352-3409 URL http://dx.doi.org/10.1016/j.dib.2026.112916

[21] Yousuf M, Alsuwian T, Amin A A, Fareed S and Hamza M 2024 *Measurement and Control* **57** 1146–1160 ISSN 0020-2940 URL http://dx.doi.org/10.1177/00202940241231473

[22] Gao Z, Jiang Z, Dong Z, Fu X, Chen Y and Zhang C 2025 *Science China Technological Sciences* **68** ISSN 1869-1900 URL http://dx.doi.org/10.1007/s11431-025-3072-9

[23] Asutkar S, Chalke C, Shivgan K and Tallur S 2023 *Expert Systems with Applications* **213** 119016 ISSN 0957-4174 URL http://dx.doi.org/10.1016/j.eswa.2022.119016

[24] Ertarğın M, Orhan A, Yıldırım Ö, Chauhan S and Vashishtha G 2026 A smartphone-based vibration dataset for induction motor fault diagnosis under different speed and load conditions URL https://data.mendeley.com/datasets/rs4vz8n3t5/1

[25] D'Emilia G and Natale E 2021 *International Journal of Metrology and Quality Engineering* **12** 15 ISSN 2107-6847 URL http://dx.doi.org/10.1051/ijmqe/2021011

[26] Shukla A, Mahmud M and Wang W 2020 *Measurement Science and Technology* **31** 105104 ISSN 1361-6501 URL http://dx.doi.org/10.1088/1361-6501/ab8dfc

[27] Prato A, Mazzoleni F, D'Emilia G, Gaspari A, Natale E and Schiavi A 2021 *Measurement* **184** 109925 ISSN 0263-2241 URL http://dx.doi.org/10.1016/j.measurement.2021.109925

[28] Schiavi A, Prato A, Mazzoleni F, Facello A and Kwitonda A 2023 Metrological characterization of digital mems accelerometers in dynamic conditions: An investigation of linearity in amplitude and temperature effects *Proceedings of the 5th IMEKO TC22 Conference on Vibration Measurement* (IMEKO) pp 1–4 URL http://dx.doi.org/10.21014/tc22-2022.033

[29] JCGM 2008 Evaluation of measurement data—guide to the expression of uncertainty in measurement (GUM) JCGM 100:2008 joint Committee for Guides in Metrology

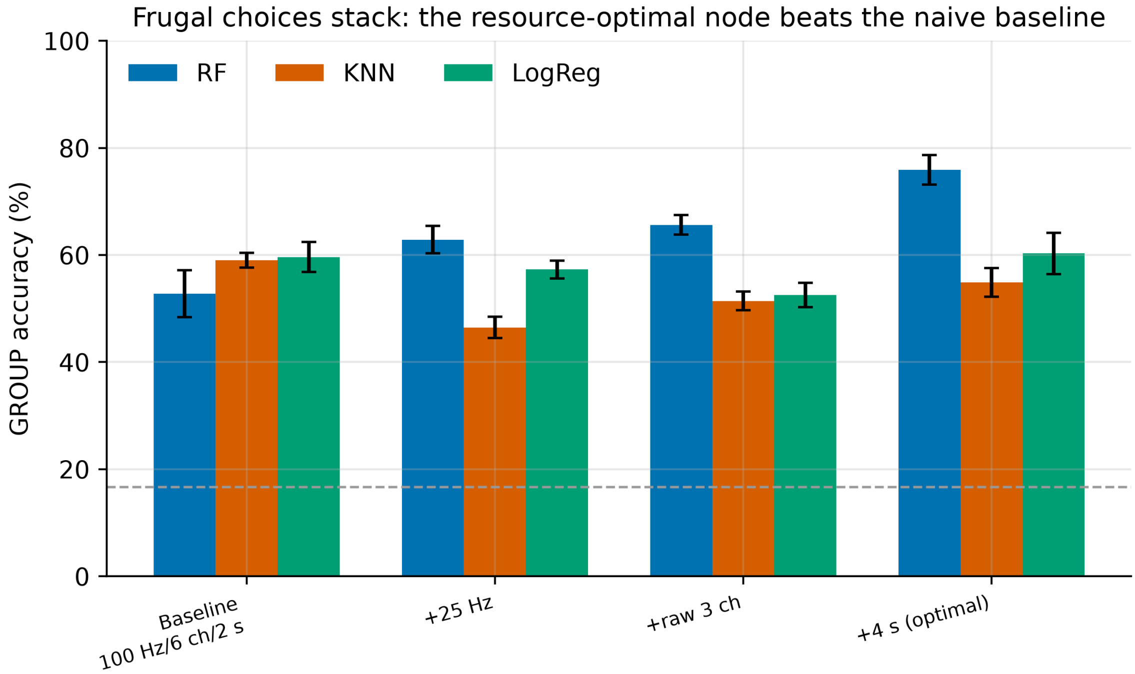

Figure 9. Incremental frugal recipe under the GROUP protocol. Combining the resource-frugal choices raises Random Forest accuracy from 52.8% to 75.9% while reducing the raw data volume roughly eight-fold.